Myntra: Cloud Scaling for Flash Sales and Personalization

Cloud Computing Case Study — Week 1 Deliverable

Sources and Salient Points

Team No: 33 | Date: 19 October 2026

1. Sources Found

ID	Source	Key information extracted
P1	Myntra Engineering (Data Platform)	Bifrost lakehouse, Quicksilver, Kafka scaling, 2.5 PB/day data serving.
P2	Myntra Engineering (Pricing System)	Discount-service latency reduced from 150 ms to approximately 30 ms.
P3	Myntra Engineering (BCP/DR)	EORS traffic spikes of 15–20×; multi-region disaster recovery.
P4	Myntra Engineering (Sourcerer)	Debezium CDC, Kafka-based ingestion and large-scale event pipelines.
V1	Microsoft (Azure Migration)	Azure migration announced in 2020; 50% higher EORS orders reported.
V2	Aerospike (Myntra Case Study)	Feature lookup 8.5 ms → 0.8 ms; up to 500K operations/sec.
V3	Databricks (Myntra Case Study)	35% lower data infrastructure cost; 25% faster real-time pipelines.

2. Case Study Salient Points

2.1 Flash-Sale Scaling and Resilience

- **Extreme traffic spikes:** Myntra's End of Reason Sale (EORS) can generate **15–20× normal traffic** (P3). Reported order-handling capacity increased from a **12,000 orders/min target in 2019** to **20,000 orders/min capacity in 2020**, while Big Fashion Festival (BFF) visits grew from approximately **460 million in 2023 to 627 million in 2024**.
- **Cloud migration:** Myntra announced its migration to **Microsoft Azure in February 2020** to support seasonal demand through scalable infrastructure. Its disaster-recovery design uses two regions with replicated data, supporting service continuity during failures (P3, V1).
- **Scalability and fault tolerance:** Load testing, independent service scaling, autoscaling, rate limiting and circuit breakers help protect services from sudden overload. Kafka capacity is increased during peak events, reducing the need for permanently provisioned resources.

2.2 Data Architecture and Personalization

- **Data processing:** Myntra's **Bifrost** data lakehouse addresses limitations of traditional warehouse architectures, while **Quicksilver** supports near-real-time event processing. Its data platform reported **2.5 PB/day** of data serving and **65,000+ queries/day** during BFF 2022 (P1).
- **Event ingestion:** Kafka-based pipelines process large volumes of customer interactions. Myntra's earlier ingestion architecture handled approximately **4 million events/minute**, while Debezium-based change data capture supports database event ingestion (P4).
- **Real-time recommendations:** Machine-learning models rank homepage widgets using customer behavior and product features. The online feature store provides low-latency access, with Aerospike reducing feature-lookup latency from **8.5 ms to 0.8 ms**, supporting up to **500,000 operations/sec** (V2).
- **Pricing optimization:** Redis read balancing and MongoDB fallback optimizations reduced discount-service latency from approximately **150 ms to 30 ms**, improving response times under heavy traffic (P2).

2.3 Cost Optimization

- **Traditional infrastructure:** Provisioning hardware for peak EORS traffic increases fixed infrastructure costs and can leave capacity underutilized during normal periods.
- **Elastic scaling:** Cloud resources can scale according to demand, while caching, efficient storage and optimized event processing reduce unnecessary compute and data-access overhead.
- **Measured improvements:** Databricks reports **35% lower data infrastructure costs** and **25% faster real-time pipelines**. Myntra's 2024 Data + AI Summit presentation separately reported **over 35% lower ingestion operating costs**. These figures refer to specific workloads, not Myntra's total cloud expenditure.

2.4 Key Outcomes

Performance metric	Reported outcome
EORS traffic	15–20× normal traffic
Discount-service latency	150 ms → approximately 30 ms
Feature-lookup latency	8.5 ms → 0.8 ms
Personalization feature operations	Up to 500K/sec
Data infrastructure cost	35% reduction
Real-time pipeline speed	25% improvement
EORS order growth	50% increase reported in 2020

Conclusion: Myntra's architecture demonstrates how cloud elasticity, distributed event processing, resilient infrastructure and low-latency caching can support flash-sale traffic while delivering personalized recommendations. The reported results highlight improvements in scalability, processing speed and infrastructure efficiency.

Note: Metrics are drawn from different years and company or partner reports. They describe specific workloads and should not be treated as a single before-and-after comparison.